User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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🔗 sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Overview

Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart

Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run

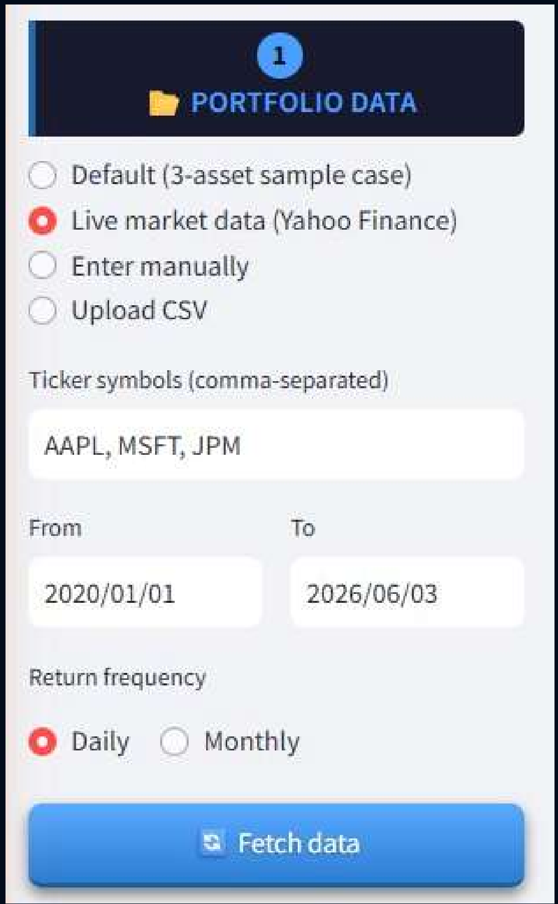
AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Three Portfolio Perspectives

1	**Portfolio (1) — No derivative:**	Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here.
2	**Portfolio (2) — With derivative:**	Same implied λ and downside constraint (H,α or H,α,L) — shows return uplift from derivatives.
3	**Portfolio (3) — Same risk as (1):**	Interpolated at P(1) std dev — direct comparison of return gain at equivalent risk (indicative).

Step 1 — Portfolio Data



Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

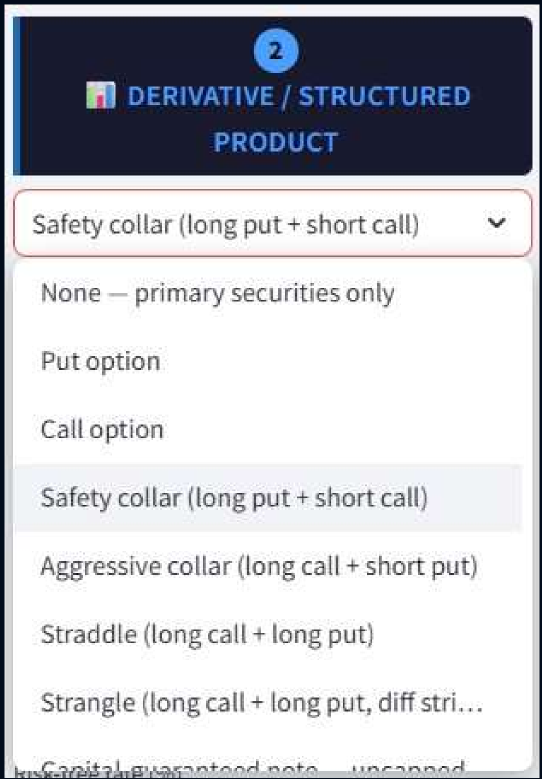
Enter manually

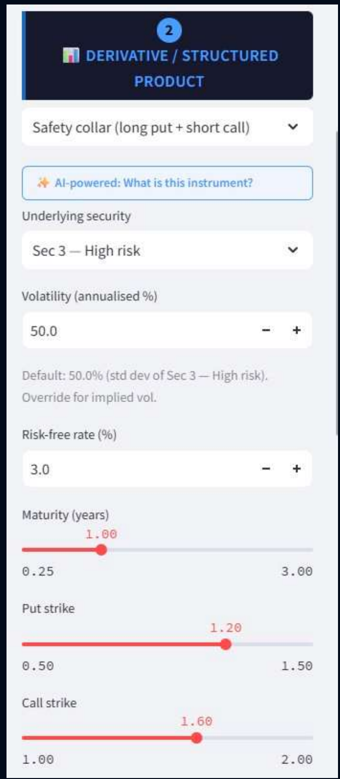
Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

2 Step 2 — Derivative / Structured Product





Available instruments

- Put option / Call option
- Safety collar (long put + short call)
- Aggressive collar (long call + short put)
- Straddle (long call + long put)
- Strangle (long call + long put, diff strikes)
- Capital-guaranteed note (capped or uncapped)
- Barrier-M note
- Custom composer (calls, puts, digitals, ZCBs)

💡 Select 'None' to optimise without derivatives — Portfolio (1) only

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Step 3 — Mental-Account Constraint

VaR — Value at Risk

$P(return < H) \leq \alpha$

The probability of the portfolio return falling below threshold H must not exceed α.
Set H (loss threshold) and α (max probability) using the sliders.

ES — Expected Shortfall

$E[return \mid return < H] \geq L$

The expected loss in the worst α% of scenarios below H must not exceed L.
Set H, α, and L using the sliders.

💡 Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

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Step 4 — Grid Resolution

⚡ Fast (m=21, m'=15)

⏱ ~1–2 min

Quick preview and parameter exploration

⚖ Standard (m=35, m'=50)

⏱ ~5–15 min

Good balance — close to thesis results

🎯 High precision (m=51, m'=99)

⏱ 1–2 hrs

Publication-quality replication of thesis results

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Step 5 — Run the Optimiser

Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

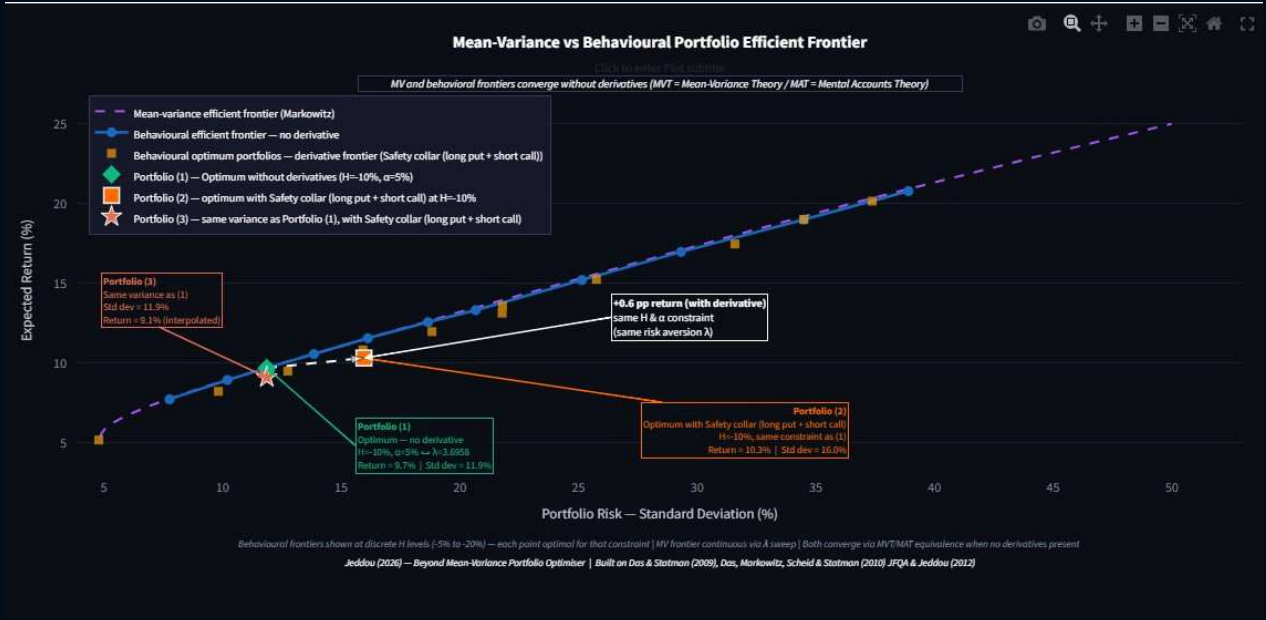


Chart legend

Purple dashed
Mean-variance efficient frontier (Markowitz)

Blue dots
Behavioural efficient frontier — no derivative

Gold squares
Behavioural optimum — derivative frontier

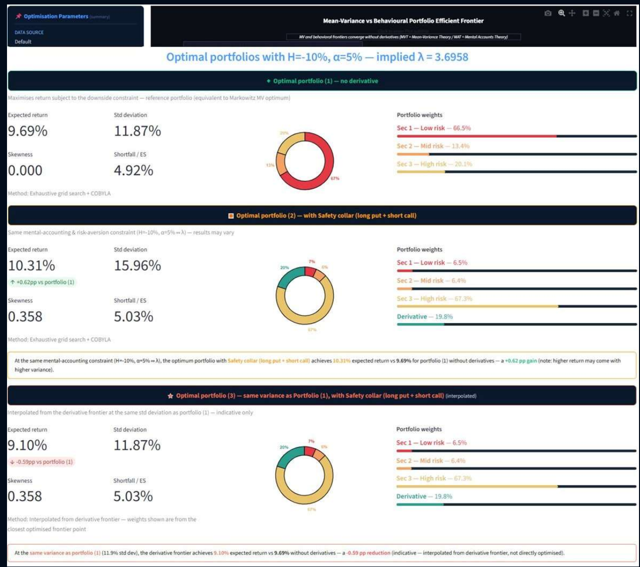
Green diamond
Portfolio (1) — no derivative optimum

Orange square
Portfolio (2) — with derivative, same λ

Coral star
Portfolio (3) — same variance as P(1)

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results



Key metrics

Expected return: Annualised mean return of the optimal portfolio

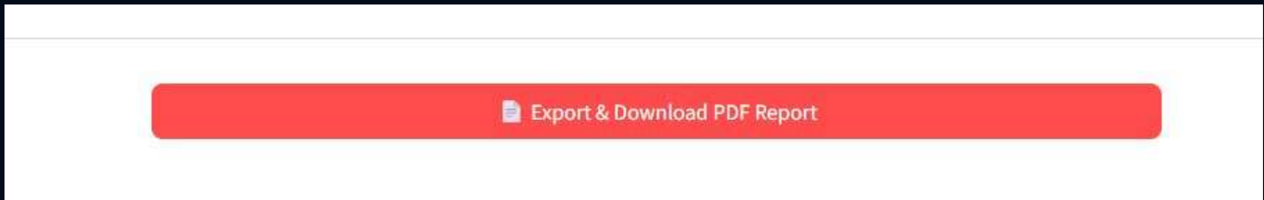
Std deviation: Annualised portfolio volatility (risk measure)

Skewness: Asymmetry of return distribution — positive = upside skew

Shortfall / ES: Expected loss in the tail (ES constraint value)

Portfolio weights: Allocation to each security and derivative

PDF Export



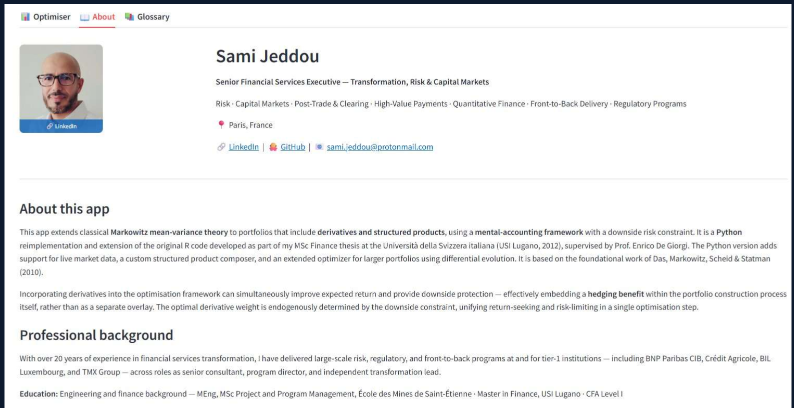
PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

💡 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

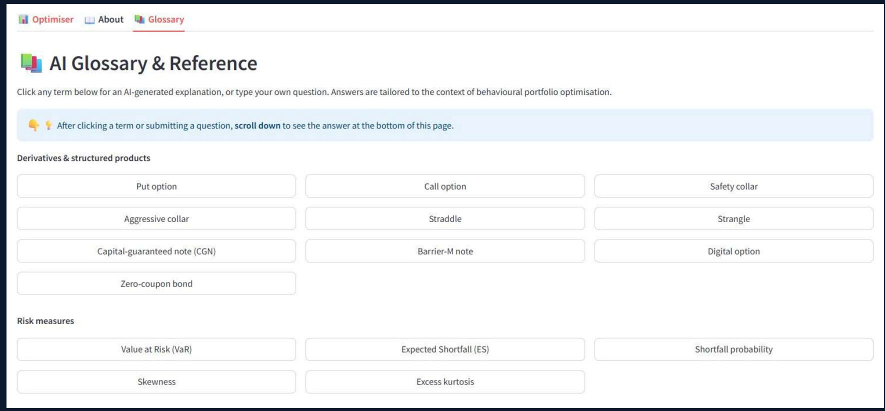
About & Glossary Tabs

📖 About Tab



- Author background & contact
- App description & algorithm
- Academic references
- Legal disclaimer

📚 Glossary Tab



- Click any term for AI explanation
- Covers derivatives, risk, portfolio theory
- Ask your own questions
- Tailored to behavioural portfolio context

Disclaimer & Contact

⚠️ Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano

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